



# Cambridge IGCSE™

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## MATHEMATICS

0580/21

Paper 2 (Extended)

May/June 2026

MARK SCHEME

Maximum Mark: 100

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Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

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This document consists of **17** printed pages.

**General marking principles**

All examiners must apply these marking principles when marking candidate responses.

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Examiners must award marks for each response in line with: <ul style="list-style-type: none"> <li>the specific <b>content</b> defined in the mark scheme</li> <li>the specific <b>skills</b> defined in the mark scheme or in the level descriptions</li> <li>the <b>standard of response</b> required as exemplified by the standardisation scripts.</li> </ul>                                                                                                                                                                                                                                                                                    |
| 2 | Examiners must award <b>whole marks</b> (not half marks, or other fractions).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3 | Examiners must award marks <b>positively</b> . This means that: <ul style="list-style-type: none"> <li>marks are not deducted for errors or omissions</li> <li>marks are awarded for correct/valid answers as defined in the mark scheme and also for <b>other valid answers</b> which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate</li> <li>the quality of spelling, punctuation and grammar are judged <b>only</b> when the mark scheme indicates that these features are <b>specifically assessed</b> in the question. However, the meaning of all responses must be unambiguous.</li> </ul> |
| 4 | Examiners must <b>consistently</b> apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5 | Examiners must use the <b>full range of marks</b> defined in the mark scheme, unless limited by the quality of the candidate responses seen.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 6 | Examiners must award marks according to the mark scheme and <b>must not</b> award marks with grade thresholds or grade descriptions in mind.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

**Mathematics-specific marking principles**

- 1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
- 2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
- 3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
- 4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
- 5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.
- 6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

**Annotations guidance for centres**

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

**Annotations**

| <b>Annotation</b>                                                                   | <b>Meaning</b>                 |
|-------------------------------------------------------------------------------------|--------------------------------|
|    | More information required      |
|    | Accuracy mark awarded zero     |
|    | Accuracy mark awarded one      |
|    | Accuracy mark awarded two      |
|  | Accuracy mark awarded three    |
|  | Independent mark awarded zero  |
|  | Independent mark awarded one   |
|  | Independent mark awarded two   |
|  | Independent mark awarded three |
|  | Benefit of the doubt           |

| Annotation       | Meaning                                                                                                                                    |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>E</b>         | Communication mark                                                                                                                         |
| <b>X</b>         | Incorrect                                                                                                                                  |
| <b>FT</b>        | Follow through                                                                                                                             |
| Highlighter      | Highlight a key point in the working                                                                                                       |
| <b>ISW</b>       | Ignore subsequent work                                                                                                                     |
| <b>M0</b>        | Method mark awarded zero                                                                                                                   |
| <b>M1</b>        | Method mark awarded one                                                                                                                    |
| <b>M2</b>        | Method mark awarded two                                                                                                                    |
| <b>M3</b>        | Method mark awarded three                                                                                                                  |
| <b>MR</b>        | Misread                                                                                                                                    |
| <b>O</b>         | Omission                                                                                                                                   |
| Off-page comment | Allows comments to be entered at the bottom of the RM marking window and then displayed when the associated question item is navigated to. |
| On-page comment  | Allows comments to be entered in speech bubbles on the candidate response.                                                                 |
| <b>Pre</b>       | Premature rounding/approximation                                                                                                           |
| <b>SC</b>        | Special case                                                                                                                               |
| <b>SEEN</b>      | Indicates that work/page has been seen                                                                                                     |
| <b>TE</b>        | Transcription error                                                                                                                        |

| Annotation                                                                        | Meaning                               |
|-----------------------------------------------------------------------------------|---------------------------------------|
|  | Correct                               |
|  | Correct answer from incorrect working |

**MARK SCHEME NOTES**

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

**Types of mark**

- M** Method marks, awarded for a valid method applied to the problem.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.
- B** Mark for a correct result or statement independent of Method marks.

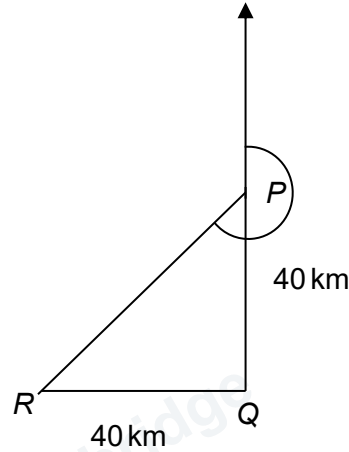
When a part of a question has two or more ‘method’ steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation ‘dep’ is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

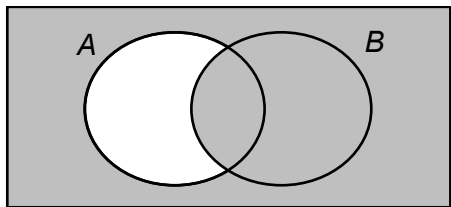
**Abbreviations**

|      |                            |
|------|----------------------------|
| awrt | answers which round to     |
| cao  | correct answer only        |
| dep  | dependent                  |
| FT   | follow through after error |
| isw  | ignore subsequent working  |
| nfw  | not from wrong working     |
| oe   | or equivalent              |
| rot  | rounded or truncated       |
| SC   | Special Case               |
| soi  | seen or implied            |

| Question | Answer             | Marks | Partial Marks                                                                                     | Guidance                                                                                                                                 |
|----------|--------------------|-------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | 1                  | 1     |                                                                                                   |                                                                                                                                          |
| 2        | 3                  | 2     | <b>M1</b> for $7x = 25 - 4$ or better                                                             |                                                                                                                                          |
| 3        | 130                | 2     | <b>B1</b> for 25 or 100 or 30<br>or <b>M1</b> for $4 \times 5^2 + 3 \times 10$                    |                                                                                                                                          |
| 4        | 30                 | 2     | <b>M1</b> for $6 = \frac{1}{12}(6 + 6 + 2L)$ oe                                                   | Allow better for <b>M1</b> , e.g. $12 \times 6 - 6 - 6$ ,<br>$72 - 6 - 6 [= 60]$ .<br>Condone 60 as answer for M1 provided working seen. |
| 5(a)     | July               | 1     |                                                                                                   |                                                                                                                                          |
| 5(b)     | 12                 | 2     | <b>B1</b> for correct vertical scale soi (minor gridlines every 3)<br>or <b>M1</b> for $9 \div 3$ | e.g. May swimming = 18<br><br>M1 may be implied by reference to one row horizontally being 3.                                            |
| 6        | $x = 5$<br>$y = 7$ | 2     | <b>B1</b> for $x = 5$<br><b>B1</b> for $y = 7$                                                    |                                                                                                                                          |



| Question | Answer                                   | Marks | Partial Marks                                                                                                                                                                                                                                                                                                                     | Guidance                                                                                                                                                                                                                                  |
|----------|------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7        | 225                                      | 2     | <b>B1</b> for $QPR$ or $QRP = 45$ or the bearing of $P$ from $R$ is 45<br>or for indicating correct bearing on diagram                                                                                                                                                                                                            | May be on diagram<br>May be implied by $180 + \text{their } RPQ$<br>or $360 - \text{their } 135$<br>Do not allow more than one bearing marked.<br><br> |
| 8(a)     | (0.5, 6) oe                              | 2     | <b>B1</b> for $(a, 6)$ , $a \neq 0.5$<br>or for $(0.5, b)$ , $b \neq 2, 6$                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                           |
| 8(b)     | $\begin{pmatrix} -1 \\ -4 \end{pmatrix}$ | 2     | <b>B1</b> for $\begin{pmatrix} k \\ -4 \end{pmatrix}$ or $\begin{pmatrix} -1 \\ k \end{pmatrix}$ or $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$                                                                                                                                                                                        | For B1, condone the vector with a fraction line.                                                                                                                                                                                          |
| 8(c)     | 2                                        | 4     | <b>B1</b> for $4 = [0^2 + a \times 0 + ] b$ soi oe<br><b>B2</b> for $a = 3$ soi<br>or <b>M1</b> for $8 = 1^2 + a[ \times 1 ] + b$ oe<br>or better e.g. with <i>their</i> $b$<br><b>M1</b> for $c = (-2)^2 + a(-2) + b$ oe<br>or better e.g. with <i>their</i> $a$ and <i>their</i> $b$<br>Maximum of 3 marks if answer incorrect. | Implied by $y = x^2 + ax + 4$<br>Implied by e.g. $y = x^2 + 3x + 4$                                                                                                                                                                       |

| Question  | Answer                                                                            | Marks | Partial Marks                                                                                                                                                                                                                                                          | Guidance                                                                                                                                                                                  |
|-----------|-----------------------------------------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9         |  | 1     |                                                                                                                                                                                                                                                                        |                                                                                                                                                                                           |
| 10(a)     | 56                                                                                | 1     |                                                                                                                                                                                                                                                                        | Allow answer $2^3 \times 7$ isw.                                                                                                                                                          |
| 10(b)     | $2^8 \times 3^8 \times 5^3 \times c^4$                                            | 2     | <b>B1</b> for $2^6 \times 3^8 \times 5^2 \times c^4$<br>or for $[20 =] 2^2 \times 5$ oe<br>or <b>M1</b> for correct factor tree/diagram/list/table for 20                                                                                                              | Allow B1 for the inclusion of 1 with correct factors.                                                                                                                                     |
| 11(a)     | $2n + 1$ oe final answer                                                          | 3     | <b>B2</b> for $3n - 4$ for sequence C<br>or <b>B1</b> for $3n + j$ or $kn - 4, k \neq 0$ for sequence C,<br>or <b>M1</b> for <i>their</i> $n$ th term $-(n - 5)$<br><br>OR<br><br><b>B2</b> for 5, 7, 9 for sequence A<br>or <b>B1</b> for $-3, -2, -1$ for sequence B | $3n - 4 - (n - 5)$ scores 3 marks.<br><br><i>their</i> $n$ th term must be equivalent to the form $an + b$ .                                                                              |
| 11(b)(i)  | 16                                                                                | 1     |                                                                                                                                                                                                                                                                        |                                                                                                                                                                                           |
| 11(b)(ii) | $2^{n-2}$ oe final answer                                                         | 2     | <b>B1</b> for $[c \times] 2^{an+k}, a > 0$<br>or for $[c \times] \left(\frac{1}{2}\right)^{k-an}, a > 0$                                                                                                                                                               | For 2 marks (or 1 if spoilt) equivalents include $\frac{2^n}{4}, \frac{1}{2}(2)^{n-1}, \left(\frac{1}{2}\right)^{2-n}$ .<br>Allow other variables for $n$ .<br>Condone e.g. $n = 2^{n-2}$ |

| Question  | Answer                            | Marks | Partial Marks                                                                                                                                                                                 | Guidance                                                                                                                                                                                               |
|-----------|-----------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12        | -5                                | 2     | <b>M1</b> for $\frac{20-10}{3-5}$ oe                                                                                                                                                          | M1 for $\frac{-5}{1}, -\frac{10}{2}$ etc.                                                                                                                                                              |
| 13(a)     | 0.71124 cao                       | 2     | <b>B1</b> for 0.92424[...] or answer 0.7112....                                                                                                                                               | Working via fractions can gain full marks if correct answer reached, or B1 for decimals as shown.<br>B1 for answer 0.71124                                                                             |
| 13(b)     | $\frac{61}{90}$ oe fraction       | 2     | <b>M1</b> for 67.77... – 6.77... oe                                                                                                                                                           | oe e.g. 6.77... – 0.67... implied by $\frac{6.1}{9}$                                                                                                                                                   |
| 14(a)(i)  | 2.5                               | 1     |                                                                                                                                                                                               | Accept answers that round to 2.5.<br>Accept 2 hours 30 minutes.<br>Do not accept 2:30 or 2.30.                                                                                                         |
| 14(a)(ii) | 1.2                               | 2     | <b>B1</b> for [LQ =] 1.8 or [UQ =] 3                                                                                                                                                          | Values may be written on diagram for B1.<br>Allow B1 if 1.8 or 3 given as answer.<br>B0 for 3 coming from 4.5 – 1.5.<br>Accept 1 hour 12 minutes.<br>Do not accept 1:12 or 1.12; these would imply B1. |
| 14(b)     | 11, 27                            | 2     | <b>B1</b> for each                                                                                                                                                                            |                                                                                                                                                                                                        |
| 15        | 6                                 | 4     | <b>M3</b> for $\frac{k}{4} = \frac{135}{360} \times 4$ oe<br>OR<br><b>M1</b> for $\frac{90}{360} \times \pi \times t^2$ oe<br><br><b>M1</b> for $\frac{135}{360} \times \pi \times (2t)^2$ oe | For M3, $\pi$ and $t^2$ should have been cancelled.<br><br>Ignore inclusion of $k$ for first two M marks<br>$\frac{1}{4}\pi t^2$ .<br><br>$\frac{3}{8} \times \pi \times 4t^2, \frac{3}{2}\pi t^2$     |
| 16        | $3x^2 + 13ax + 4a^2$ final answer | 2     | <b>B1</b> for three correct terms from $3x^2 + ax + 12ax + 4a^2$                                                                                                                              | $13ax$ counts as two correct terms.                                                                                                                                                                    |

| Question | Answer                                           | Marks | Partial Marks                                                                                                                                                                                                 | Guidance                                                                                                                                                                                                                                    |
|----------|--------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17(a)    | Shape drawn at (0, 7), (−4, 9), (−6, 5), (−2, 5) | 2     | <b>B1</b> for reflection in $x = k$ or in $y = 1$ or for three correct vertices                                                                                                                               | Note: reflection in $y = 1$ has coordinates (2, −5), (4, −3), (8, −3), (6, −7).                                                                                                                                                             |
| 17(b)    | Enlargement<br>$-\frac{1}{2}$<br>(2, −1)         | 3     | <b>B1</b> for each                                                                                                                                                                                            | Do not allow dilation or reduction.<br>If more than one transformation, award 0 marks.<br>Treat extra properties as choice.<br>Condone 2, −1 and ‘origin 2, −1’.<br>Do not treat centre as a column vector as an additional transformation. |
| 18       | $\frac{1}{2}$ oe                                 | 2     | <b>B1</b> for $\sin 60 = \frac{\sqrt{3}}{2}$ or $\tan 30 = \frac{1}{\sqrt{3}}$ oe                                                                                                                             | isw simplification of fraction not containing surds.<br><b>B1</b> for e.g. $\frac{\sqrt{3}}{2\sqrt{3}}$ or for $\frac{\sqrt{3}}{2} \times a$ or for $b \times \frac{1}{\sqrt{3}}$<br>Do not pick <b>B1</b> out of a table of values.        |
| 19       | [a =] 5<br>[b =] −2                              | 2     | <b>B1</b> for each                                                                                                                                                                                            |                                                                                                                                                                                                                                             |
| 20       | 4                                                | 4     | <b>B3</b> for $\frac{1}{9}$ and 36<br>or $3^{-2} \times 3^2 \times 2^2$<br>or <b>B2</b> for $\frac{1}{9}$ or 36<br>or $3^{-2} \times 6^2$ or $2^2 \times 3^2$<br>or <b>B1</b> for $\frac{1}{3^2}$<br>or $6^2$ |                                                                                                                                                                                                                                             |
| 21(a)    | 108                                              | 1     |                                                                                                                                                                                                               |                                                                                                                                                                                                                                             |
| 21(b)    | 168                                              | 3     | <b>M2</b> for [2×] ( $2 \times 6 + 2 \times 9 + 6 \times 9$ )<br>or <b>M1</b> for $2 \times 6$ or $2 \times 9$ or $6 \times 9$                                                                                |                                                                                                                                                                                                                                             |

| Question | Answer         | Marks | Partial Marks                                                                                                                                                                                                                                                                                        | Guidance                                                                                                                                                                                                                                                                                                                            |
|----------|----------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21(c)    | 11             | 3     | <b>M2</b> for $2^2 + 6^2 + 9^2$ oe or better<br>or <b>M1</b> for $2^2 + 6^2$ or $2^2 + 9^2$ or $6^2 + 9^2$ oe or better                                                                                                                                                                              | M2 for 121 or $\sqrt{121}$<br>M1 for 40 or $\sqrt{40}$ or 85 or $\sqrt{85}$ or 117 or $\sqrt{117}$                                                                                                                                                                                                                                  |
| 21(d)    | $\frac{6}{11}$ | 2     | <b>FT</b> $\frac{6}{\text{their (c)}}$<br><b>M1</b> for recognising correct angle                                                                                                                                                                                                                    | Do not isw.<br>May be on diagram.<br>M1 implied by $\sin^{-1}\left(\frac{6}{\text{their (c)}}\right)$ or<br>$\sin\left(\frac{6}{\text{their (c)}}\right)$                                                                                                                                                                           |
| 22       | 7 6 10         | 3     | <b>B2</b> for 2 correct<br>or <b>B1</b> for 1 correct<br>or <b>M1</b> for [2], 3.5, 3 and 5 oe seen [FDs]                                                                                                                                                                                            | Allow e.g. $\frac{35}{10}$ , $\frac{45}{15}$ and $\frac{25}{5}$ for M1.                                                                                                                                                                                                                                                             |
| 23       | 7.5 oe         | 3     | <b>M2</b> for $h = \sqrt[3]{\frac{27}{64}} \times 10$ oe or $\frac{10}{h} = \frac{4}{3}$ oe<br><br>or <b>M1</b> for $\sqrt[3]{\frac{27}{64}}$ or $\sqrt[3]{\frac{64}{27}}$ oe or better<br><br>or for $\left(\frac{10}{h}\right)^3 = \frac{64}{27}$ or $\left(\frac{h}{10}\right)^3 = \frac{27}{64}$ | oe e.g. $\frac{30}{4}$ , $h = \frac{3}{4} \times 10$ , $10 \div \frac{4}{3}$ , $10 \div \sqrt[3]{\frac{64}{27}}$ ,<br>$10 \times \sqrt[3]{\frac{27}{64}}$ , $\sqrt[3]{1000 \times \frac{27}{64}}$<br><br>e.g. M1 for [L]SF = $\frac{3}{4}$ or 0.75 but not just $\frac{3}{4}$ or 0.75<br><br>e.g. $h^3 = \frac{27}{64} \times 10^3$ |

| Question | Answer             | Marks | Partial Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Guidance                                                                                                                                                                                                                                            |
|----------|--------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 24(a)    | (0, 7)<br>(4, -25) | 6     | <p><b>B2</b> for <math>3x^2 - 12x</math></p> <p>or <b>M1</b> for intention to find derivative so<br/>by <math>3x^2 + kx</math> or <math>kx^2 - 12x</math><br/>or <math>3x^2 - 12x + 7</math></p> <p><b>M1</b> for setting <i>their</i> <math>\frac{dy}{dx} = 0</math></p> <p>or stating <math>\frac{dy}{dx} = 0</math></p> <p><b>M1</b> for correct method to solve <i>their</i><br/>quadratic, e.g. <math>[3]x(x - 4)</math> oe<br/>or <math>\frac{- -12 \pm \sqrt{(-12)^2 - 4(3)(0)}}{2(3)}</math> oe</p> <p><b>B1</b> for (4, -25) or for <math>x = 0</math> and <math>x = 4</math> soi</p> | <p>isw second derivative.<br/>Condone poor notation for derivative.<br/>e.g. B2 for <math>y = 3x^2 - 12x</math>, similarly for M1.</p> <p><i>their</i> <math>\frac{dy}{dx}</math> must not be second derivative.</p> <p><math>x(3x - 12)</math></p> |

| Question  | Answer                                                                                                                                                               | Marks | Partial Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Guidance                                                                                                                                                                                                                     |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 24(b)     | <p>A correct sketch with<br/>(0, 7) maximum stated<br/>(4, -25) minimum stated</p>  | 3     | <p><b>B2</b> for a correct sketch with<br/>(0, 7) maximum stated or<br/>(4, -25) minimum stated<br/>or for a correct sketch with appropriate<br/>points labelled as maximum and<br/>minimum<br/>or for a correct sketch with (0, 7)<br/>and (4, -25) marked at the maximum and<br/>minimum points</p> <p>or <b>M1</b> for reasonable sketch of any cubic</p> <p>or <b>B1</b> for a sketch not of cubic shape with<br/>identification of local maximum at (0,7)<br/>and a local minimum at (4,-25)</p> | <p>For M1, cubic may be positive or negative.<br/>Cubic should have only two turning points.<br/>Ignore position of any horizontal and / or<br/>vertical axes drawn.</p> <p>0 marks if no sketch drawn.</p>                  |
| 25(a)(i)  | $5n - m$                                                                                                                                                             | 1     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                              |
| 25(a)(ii) | $2n - \frac{1}{5}m$ or $\frac{1}{5}(10n - m)$                                                                                                                        | 3     | <p><b>M2</b> for <math>\frac{1}{5}m + \frac{2}{5}</math> their <math>(5n - m)</math> oe<br/>or <b>M1</b> for a correct route</p> <p>or for <math>\overline{BQ} = \frac{2}{5}</math> their <math>(5n - m)</math></p> <p>or for <math>\overline{CQ} = -\frac{3}{5}</math> their <math>(5n - m)</math></p>                                                                                                                                                                                               | <p><math>-\frac{4}{5}m + 5n - \frac{3}{5}</math> their <math>(5n - m)</math><br/>e.g. <math>PB + BQ</math><br/>Condone a mix of notation<br/>or <math>\overline{QC}</math> or <math>\overline{QB}</math> correct for M1.</p> |

| Question | Answer                                                                                                                                                                                                                                                            | Marks     | Partial Marks | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 25(b)    | $\overrightarrow{QD} = 6\mathbf{n} - \frac{3}{5}\mathbf{m}$ oe<br>or $\overrightarrow{PD} = 8\mathbf{n} - \frac{4}{5}\mathbf{m}$ oe                                                                                                                               | <b>B1</b> |               | Allow unsimplified                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|          | For showing any two of $\overrightarrow{QD}$ , $\overrightarrow{PD}$ and $\overrightarrow{PQ}$ share a common factor, e.g.<br>$\overrightarrow{QD} = 3(2\mathbf{n} - \frac{1}{5}\mathbf{m})$<br>or $\overrightarrow{PD} = 4(2\mathbf{n} - \frac{1}{5}\mathbf{m})$ | <b>B1</b> |               | Other pairs with common factors are acceptable<br>e.g. $\overrightarrow{PQ} = 2\mathbf{n} - \frac{1}{5}\mathbf{m}$ and $\overrightarrow{QD} = 3(2\mathbf{n} - \frac{1}{5}\mathbf{m})$<br>$\overrightarrow{PD} = \frac{4}{5}(10\mathbf{n} - \mathbf{m})$ and $\overrightarrow{PQ} = \frac{1}{5}(10\mathbf{n} - \mathbf{m})$<br>If factorisation not shown, allow correct vectors in terms of $\mathbf{m}$ and $\mathbf{n}$ followed by a correct relationship, e.g.<br>$\overrightarrow{QD} = \frac{3}{4}\overrightarrow{PD}$ , $4\overrightarrow{QD} = 3\overrightarrow{PD}$ , $\overrightarrow{QD} : \overrightarrow{PD} = 3 : 4$ ,<br>$\frac{\overrightarrow{QD}}{\overrightarrow{PD}} = \frac{3}{4}$<br>$\overrightarrow{PQ} = \frac{1}{4}\overrightarrow{PD}$ , $4\overrightarrow{PQ} = \overrightarrow{PD}$ , $\overrightarrow{PQ} : \overrightarrow{PD} = 1 : 4$ ,<br>$\frac{\overrightarrow{PQ}}{\overrightarrow{PD}} = \frac{1}{4}$<br>$\overrightarrow{QD} = 3\overrightarrow{PQ}$ , $\frac{1}{3}\overrightarrow{QD} = \overrightarrow{PQ}$ , $\overrightarrow{QD} : \overrightarrow{PQ} = 3 : 1$ ,<br>$\frac{\overrightarrow{QD}}{\overrightarrow{PQ}} = 3$ |
| 25(b)    | A correct conclusion, e.g.<br>$\overrightarrow{QD}$ is a multiple of/is parallel to $\overrightarrow{PQ}$<br>and they have a common point $Q$ , so<br>they are collinear oe                                                                                       | <b>B1</b> |               | e.g. with common point $P$ if $\overrightarrow{PQ}$ and $\overrightarrow{PD}$ used,<br>or with common point $D$ if $\overrightarrow{QD}$ and $\overrightarrow{PD}$ used.<br><br>Allow without vector arrows for all B1 marks in <b>25(b)</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |



| Question | Answer               | Marks | Partial Marks                                                                                                                                                                                                                                                                                                                                                                               | Guidance                                                                                                                                                                                                                                                                                                                                                                              |
|----------|----------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 26(a)    | $11 - 7\sqrt{2}$ cao | 3     | <b>B2</b> for $11 + 3\sqrt{2} - 5\sqrt{8}$<br>OR<br><b>M1</b> for $15 + 3\sqrt{2} - 5\sqrt{8} - \sqrt{2 \times 8}$ oe with at least three terms correct.<br><b>M1</b> for $\sqrt{8} = 2\sqrt{2}$ soi                                                                                                                                                                                        | 11 or $-7\sqrt{2}$ each count as two terms correct<br>e.g. $5\sqrt{8} = 10\sqrt{2}$                                                                                                                                                                                                                                                                                                   |
| 26(b)    | $[k =] 1.9$ oe       | 4     | <b>B3</b> for $\frac{19}{2\sqrt{5}}$ oe<br>or <b>M3</b> for $\frac{7\sqrt{5}}{10} + \frac{12\sqrt{5}}{10}$ oe rational common denominator<br>OR<br><b>B1</b> for $\sqrt{\frac{49}{20}}$ or better<br><br><b>M1</b> for $\frac{7}{2\sqrt{5}}$ and $\frac{12}{2\sqrt{5}}$ oe common denominator (not rational)<br>or $\frac{7\sqrt{5}}{10}$ and $\frac{6\sqrt{5}}{5}$ oe rational denominator | B3 for answer $1.9\sqrt{5}$<br>or better for M3, e.g. $\frac{7}{2} + 6 = 5k$<br><br>Note: the root symbol should extend below the fraction line.<br>or better for B1 e.g. $\frac{7}{2\sqrt{5}}$ , $\frac{7}{\sqrt{20}}$<br>oe e.g. $\frac{7}{\sqrt{20}}$ and $\frac{12}{\sqrt{20}}$ , $7 + 12 = (k\sqrt{5})(2\sqrt{5})$<br>oe e.g. $\frac{7\sqrt{5}}{10}$ and $\frac{6\sqrt{20}}{10}$ |